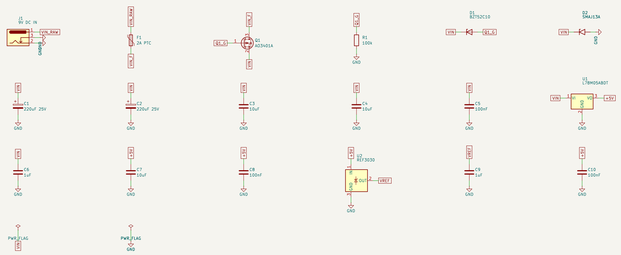
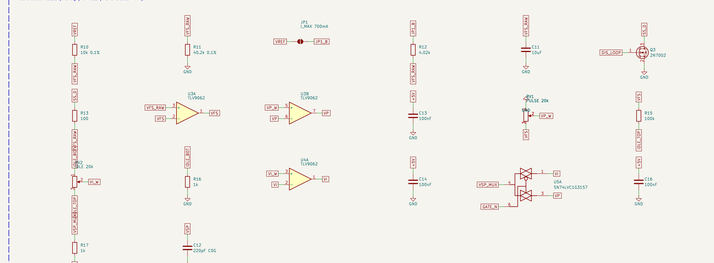
1 POWER INPUT, 5 V AND 3.000 V REFERENCE

9 V / −+5.5 A desktop adapter (5.5/2.1 mm, centre +).
Q1 = reverse-polarity protection, F1 = resettable fuse, D2 = TVS.



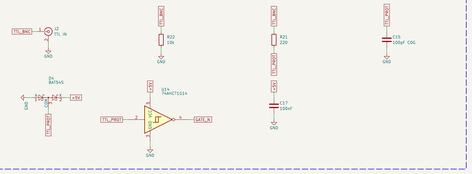
2 SETPOINT, SOFT START AND TTL GATING (scale 4 V/A)

VFS = 2.40 V = 600 mA full scale (2P1 closed; 2.80 V = 700 mA).
C11 + R12|R13 give an 80 ms soft start; Q3 discharges it when disabled.
U5 selects PULSE (TTL high) or IDLE (TTL low, 5.10k mA).



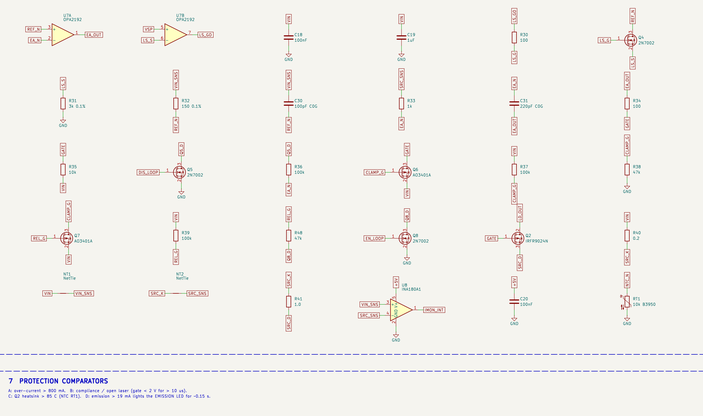
3 TTL INPUT (BNC, 10 k to GND, 3.3/5 V logic)

Schmitt trigger with TTL thresholds; unconnected input = IDLE.



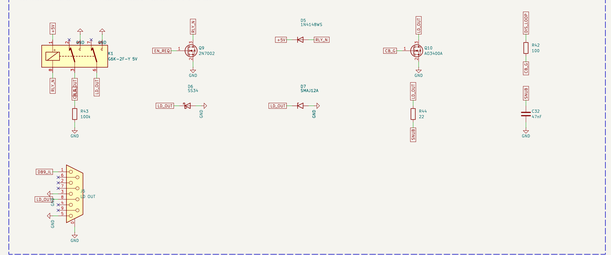
4 HIGH-SIDE CURRENT SOURCE (laser cathode at GND)

LLD = VSP × (R32/R31) // Rhd = VSP / ≈ V/A.
U7A error amp senses Rhd directly (Kelvin via NTL/M12); Rhd degenerates Q2.
Q6 clamps Q2 off unless EN_LOOP is high (fail-safe; QT/QR release 5).
Q5 forces U7A high while disabled so the integrator cannot wind up.



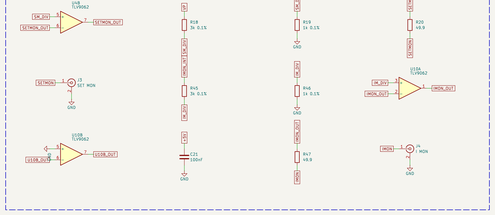
5 OUTPUT, SHORTING RELAY AND CROWBAR

K1 NC contacts short the laser whenever the driver is not armed or unpowered.
Q10 crowbar shunts 5 within 1 us of a fault; D8/D7 reverse and surge clamp.
D99 follows Thorlabs LD-driver 'CC' pinout: 8 = anode, 3 = cathode/GND, 1–5 interlocks.



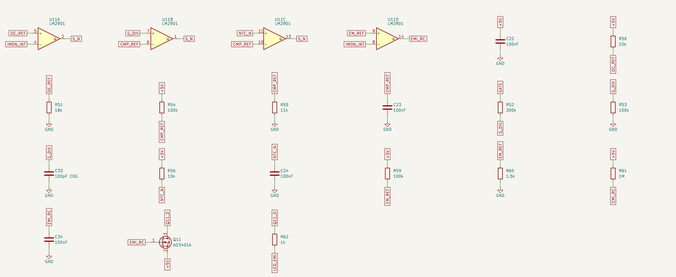
6 CURRENT AND SETPOINT MONITORS (1 V/A into high-Z)

I MON from INA180 (4 V/A) // 4; SET MON shows the PULSE setpoint // 4.



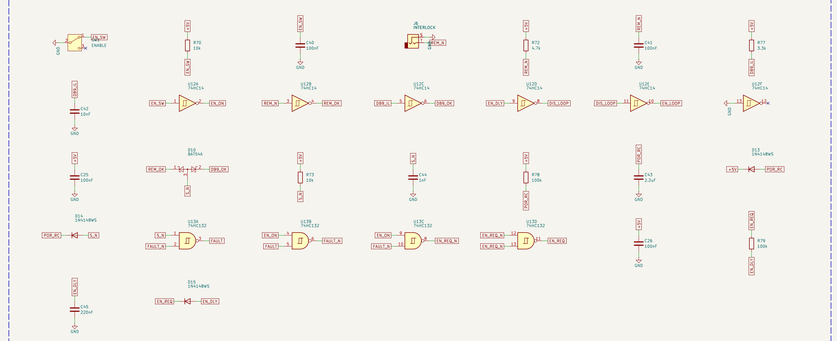
7 PROTECTION COMPARATORS

A over-current = 800 mA. B: compliance / open laser (gate = 2 V for > 10 us).
C: Q2 heatsink > 85 C (NTC M11). D: enclosure > 19 mA (gets the EMISSION LED for ~0.15 s.

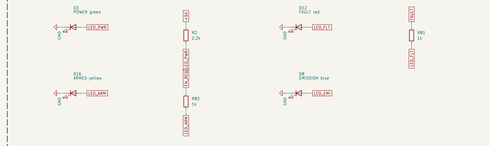


8 INTERLOCKS, FAULT LATCH AND ENABLE SEQUENCING

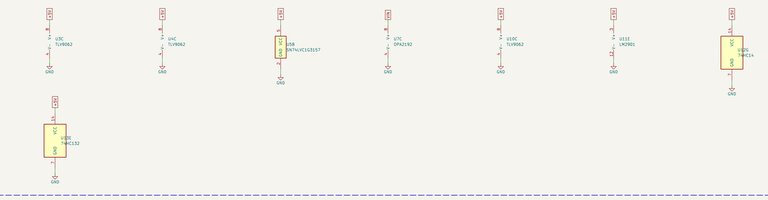
Any fault, open interlock or power-up sets the FAULT latch; reset by switching ENABLE off.
EN_REQ arms relay K1 at once; EN_LOOP releases the current source ~17 ms later.



9 FRONT-PANEL INDICATORS



10 IC POWER PINS



11 MOUNTING HOLES



CLASS 4 LASER — read README safety section before use
Laser: DTR BDR-209 405 nm (500 mA nominal, 600 mA max), CATHODE AT GND
Generated from tools/design.py by tools/gen_sch.py
ion-trap lab photoionisation 405 nm driver
Sheet: /
File: 405-driver.kicad_sch

Title: 405 nm laser diode pulse driver (BDR-209)

Size: A2	Date: 2026-10-28
KiCad 10.0.x 9.0.9	Rev: 1.0
	ID: 1/1